Priyanka Bose

Doctoral Researcher in Materials Chemistry
priyanka.bose@example.ac.in | New Delhi, India
orcid.org/0000-0001-7734-2210

Summary

Materials chemist studying perovskite stability for low-cost solar cells, with laboratory teaching experience.

EDUCATION:

2021–2026 (Expected) **Ph.D., Chemistry** — Indian Institute of Technology Delhi
2019–2021 **M.Sc., Chemistry** — University of Delhi (CGPA 9.1/10)
2016–2019 **B.Sc. (Hons), Chemistry** — University of Delhi (84.6%)

TEACHING AND RESEARCH EXPERIENCE:

2021–Present | **Teaching Assistant** | Indian Institute of Technology Delhi
Run the undergraduate physical chemistry laboratory for 80 students each semester.
Prepare and grade weekly laboratory reports.
Supervised four undergraduate research projects on thin-film deposition.
Wrote the laboratory safety manual adopted by the department in 2023.
Coordinated instrument booking for the shared Raman facility.
2020–2020 | **Summer Research Fellow** | Council of Scientific and Industrial Research
Synthesised lead-free perovskite films and measured their band gaps.
Characterised film morphology with scanning electron microscopy.
Presented results at the institute summer symposium.
Science Tutor | Pratham Education Foundation
Taught weekend chemistry classes to secondary school students.

PUBLICATIONS:

[1] Moisture-resistant tin perovskites through bulky cation passivation, ACS Energy Letters, 2024
[2] In situ Raman tracking of perovskite degradation under illumination, Journal of Physical Chemistry C, 2023
[3] A low-temperature route to stable perovskite thin films, Indian Patent Application 202311009876, 2023
[4] Halide segregation in mixed tin-lead perovskites under bias, Chemistry of Materials, 2022
[5] Solution-processed hole transport layers for flexible solar cells, Materials Today Energy, 2022
[6] Thermal stability of formamidinium tin iodide films, RSC Advances, 2021
[7] Antisolvent engineering for pinhole-free tin perovskite layers, Solar RRL, 2024
[8] Role of excess tin fluoride in suppressing oxidation, ACS Applied Energy Materials, 2023
[9] Grain boundary passivation with zwitterionic additives, Nanoscale, 2023
[10] A review of lead-free perovskite photovoltaics for Indian climates, Current Science, 2022

FELLOWSHIPS AND AWARDS:

Prime Minister's Research Fellowship, Ministry of Education, 2021
INSPIRE Scholarship for Higher Education, Department of Science and Technology, 2016

CONFERENCE PRESENTATIONS:

Oral talk, Materials Research Society of India Annual Meeting 2024; Poster, International Conference on Perovskite Photovoltaics 2023; Poster, Chemical Research Society of India Symposium 2022

LABORATORY AND COMPUTATIONAL SKILLS:

Characterisation – Raman spectroscopy | X-ray diffraction | UV-Vis spectroscopy
Computation – Python | Origin | Quantum ESPRESSO

MEMBERSHIPS:

Member, Royal Society of Chemistry; Student Member, Materials Research Society of India

LANGUAGES:

Bengali (Native), English (Fluent), Hindi (Fluent)

OUTREACH:

2022–Present | **Science Communicator** | Vigyan Prasar

Write monthly explainers on renewable energy for school students.

Host live demonstrations at the annual science festival.

2016–2019 | **Volunteer** | National Service Scheme, University of Delhi

Organised adult literacy sessions in two villages near Delhi.